ISOTOP

SINCE 2026

Concept : Neutralizing Uranium energy for specialized research and technical applications, supporting Green, Blue, and Circular economies.

Implementation :

Phase 1 (Geospatial Intelligence): Generating thermal maps via satellite imagery for precise Uranium localization.

Phase 2 (Dynamic Control): Implementing an AI-driven Reactor Manageme System ○ .prioritized by real-time demand

Challenges & Gaps

1. Cities and industrial sectors suffer from a massive shortage of 24/7 baseload power that can operate without harmful chemical emissions, hindering the transition to true zero carbon energy.
2. Current uranium mining relies on random drilling, causing irreversible damage to ecosystems and the permanent loss of green spaces.
3. Existing energy systems fail to coordinate in real time between supply and demand this leads to massive heat and electricity waste due to the absence of a dynamic link between water desalination, hydrogen production, and industrial needs.

Competitive Landscape

NuScale Power & Oklo Inc

Specialized in building SMRs and Micro-reactors for data centers and cities, but they lack the integrated software for dynamic energy management.

TerraNoX AI & Earth AI

Leading startups using AI for geological data analysis to find Uranium, but they do not handle energy consumption or reactor management.

Cameco & Kazatomprom

Global leaders in Uranium production, relying on massive traditional mining methods. Their adoption of AI is still slow and limited.

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Solution

1. We utilize Thermal Mapping and Spectral Data to locate Uranium deposits with metric precision. This eliminates the need for random drilling, preserving vegetation and protecting ecosystems from irreversible damage.
2. Our intelligent system dynamically synchronizes Uranium energy with three vital resources:
   - Water Desalination: Powering desalination plants at peakdemand hours
   - Hydrogen Production: Converting surplus energy into clean hydrogen fuel
   - Circular Economy: Recycling waste and capturing thermal energy from mineral processing for maximum efficiency.
3. By implementing Reinforcement Learning (RL), we monitor nuclear fuel status in real-time. This optimizes consumption to ensure the Nuclear Battery remains stable and operational for 20 years without human intervention.

What sets us apart from the competition?

We are the only firm bridging the gap between Smart Exploration (finding resources) and Real-Time Energy Management in one seamless ecosystem.

By focusing on Software-as-a-Solution (SaaS) and remaining politically neutral, we fast-track international licensing and gain global public acceptance

Our system doesn't just monitor; it predicts failures before they occur using advanced AI, ensuring zero downtime

We don't just sell electricity. By redirecting waste heat, we boost Uranium utilization efficiency to 40%, creating a truly circular energy model.

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Our Technical Moat

- **Quantum Chemistry Simulation:** Simulating atomic level reactions to develop ultra-precise materials for extracting Uranium from seawater.
- **AI-Quantum RL :** Utilizing Quantum-enhanced Reinforcement Learning to predict neutron behavior with accuracy that far surpasses classical computing.

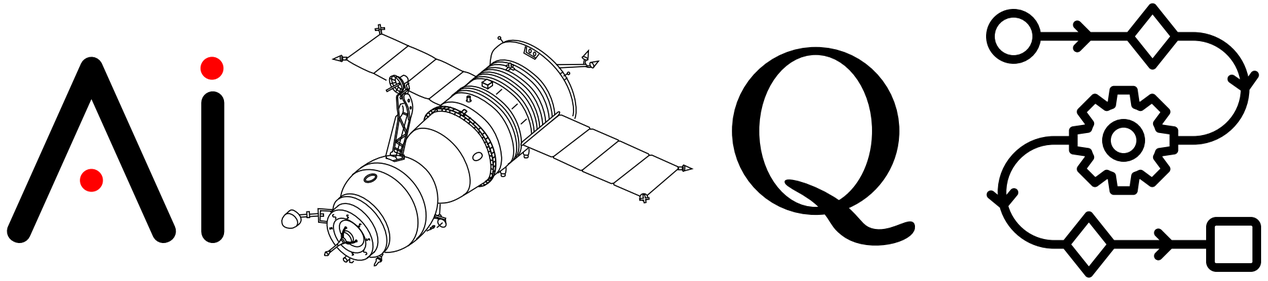
Why they can't copy us?

- Competitors rely on outdated physical models; we own proprietary quantum simulations that require years of mathematical development to replicate.
- Our system is protected by encryption protocols resistant to quantum attacks, ensuring absolute energy data security.

Current Status & Scalability

- Real-time on-site processing for security, with cloud-based analytics for global updates.
- Stage: **MVP** (Proof of Concept). Engineering & technical blueprints are complete; currently developing the first software prototype.

Technical Stack



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BUSINESS MODEL

Total Addressable Market (TAM)

- **Unit Economics**: $5 Million/Year (AI & Management licensing per SMR unit).
- **Market Projection**: With 300 Small Modular Reactors (SMRs) planned globally by 2030.
- **Total Target Market**: $1.5 Billion Annual Recurring Revenue (ARR) in the software sector alone

ISOTOP Strategic Positioning

- **For Mining Corporations**: Reducing exploration costs via high-precision satellite mapping.
- **For Power Plants**: Extending fuel life-cycles through an autonomous smart OS.
- **For Sustainable Cities**: Guaranteeing zero-waste energy distribution for desalination and Hydrogen

Current Market Spend

- **Operational Waste**: Energy firms spend 15-20% of budgets on legacy fuel management and manual maintenance.
- **Inefficient Exploration**: Billions are lost annually on "blind drilling" due to inaccurate geological mapping.
- **The Opportunity**: With $1.1 Trillion invested in low-carbon solutions last year, the market is ready to pay for efficiency-driven AI technologies.

MARKET ENTRY

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Market Entry & Customer Acquisition (CAC)

- **Pilot Projects: Partnering with National Nuclear Authorities and Desalination Giants to build trust.**
- **Global Presence: Leveraging International Energy Forums (e.g., IAEA) to mitigate long sales cycles and high initial acquisition costs through direct industry networking**

Revenue Model & Lifetime Value (LTV)

- **The "Double-Revenue" Model:**

1. **Subscription: Annual licensing fee for the AI Management Platform.**
2. **Performance Fee: 5% share of the total fuel costs saved via AI optimization.**

- **LTV: Extremely high retention, with 20-year contracts ensuring long-term recurring revenue.**

Pilot to Contract

- **Providing thermal mapping reports to prove resource potential.**
- **Running our system on the client's real data to showcase expected efficiency.**
- **Signing the long-term operational contract after proven KPI success.**

The Unfair Advantage

- The world's most precise system, merging Quantum computing with Uranium management
- **Establishing the "Gold Standard"; once the first reactor succeeds, it creates a competitive necessity for all regional reactors to adopt ISOTOP to remain cost-competitive**

EXECUTION PLAN

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Timeline

- **Q2 2026 (Current): Finalizing technical logic and publishing theoretical frameworks for the "Isotopes-to-Energy" bridge.**
- **Q3 2026: Developing the End-to-End Application prototype and establishing core Safety & Intellectual Property (IP)**
- **Month 12: Securing the first Strategic Partnership with a National Nuclear Authority or a major Desalination firm.**
- **Month 24: Full launch of the Enterprise Version and expanding operations into Seawater Uranium Extraction.**

Startup Execution

- **Month 1 (MVP Development): Building the first control panel (Dashboard) to visualize s results. Calibrating RL algorithms to target a 20% efficiency boost**
- **.Month 2 (IP & Data Validation): Initiating Patent Filings for core algorithms. Collaborat nuclear and satellite experts to integrate real-world data for system refinement.**
- **Month 3 (Market Traction): Presenting the prototype to the first potential client to sec Intent (LOI). Launching the Pre-Seed funding round to expand the engineering team**

VISIONARY FOUNDER

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Sarah Ahmed Abumandil | Founder

Academic Excellence :

- **Final-year AI & Data Science student, specializing in complex algorithmic simulations.**
- **Dental Surgery Candidate, focusing on the cognitive integration between Medical Sciences and Artificial Intelligence.**

Track Record & Innovation :

- **Founder of Dentiligence, a pioneer brand merging Dental expertise with Digital Intelligence.**
- **Distinguished participant in high-stakes hackathons, including the Palestine Quantum Computing & AI Hackathon, specializing in Quantum-Classical hybrid models.**

Any Questions